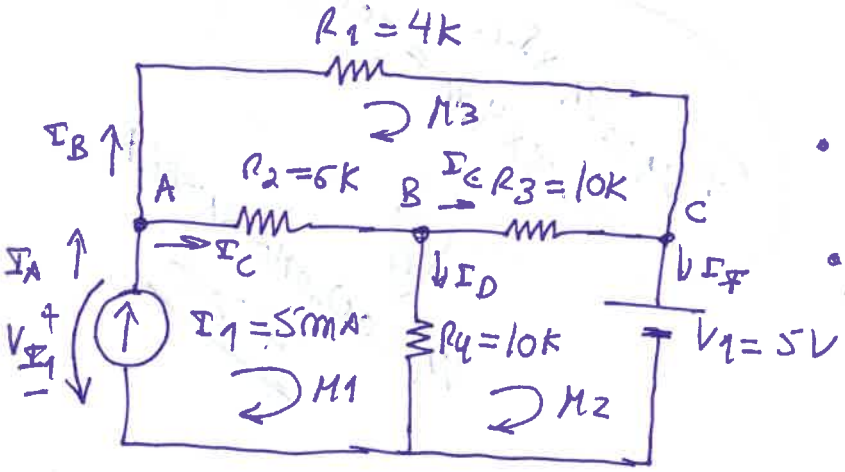
FICHA PRÁTICA 3. AULA 4

8/a) $P_{V_{I_1}} = ?$ (MÉTODO DIRETO)



- $P_{V_{I1}} = V_{I1} \times (-I_A) < 0 \Rightarrow$ FONTE
- $P_{V_{I1}} = V_{I1} \times (-I_A) > 0 \Rightarrow$ CARGA

Método direto $\Rightarrow$ N − 1 + M equações = 4 − 1 + 3 = 6 equações.

Nó A: $I_A = I_B + I_C$

Nó B: $I_C = I_E + I_D$

Nó C: $I_E + I_B = I_F$

M1: $-V_{I1} + R_2 \cdot I_C + R_4 I_D = 0$

M2: $-R_4 I_D + R_3 \cdot I_E + V_1 = 0$

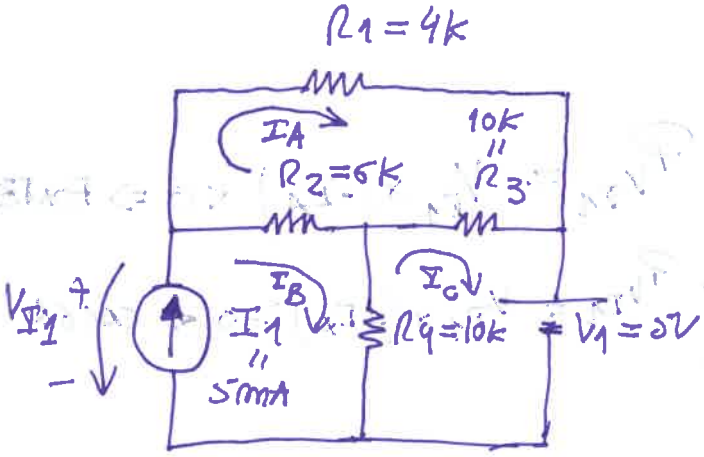
M3: $R_1 \cdot I_B - R_3 \cdot I_E - R_2 \cdot I_C = 0$

Como $I_A = I_1$, é possível introduzir como nova incógnita. $\Rightarrow V_{I_1} = ?$

~~I_A~~ $\Rightarrow V_{I1}$

~~I_A~~	I_B	I_C	I_D	I_E	I_F	V_{I1}	
I_A	$-I_B$	$-I_C$	0	0	0	0	$= 0$
0	0	I_C	~~$-I_D$~~	$-I_E$	0	0	$= 0$
0	I_B	0	0	$-I_E$	$-I_F$	0	$= 0$
0	0	$R_2 I_C$	$R_4 I_D$	0	0	$-V_{I1}$	$= 0$
0	0	0	$-R_4 I_D$	$R_3 I_E$	0	0	$= -V_1$
0	$R_1 I_B$	$-R_2 I_C$	0	$-R_3 I_E$	0	0	$= 0$

8 b)



$P_{V1} = V_1 \times I_C = 5 \times 4m = 20mW$

V_1 consome energia

$P > 0$

$$\begin{cases} R_1 \cdot I_A + R_3 \cdot (I_A - I_C) + R_2 \cdot (I_A - I_B) = 0 & (\text{malha } A) \\ R_3 \cdot (I_C - I_A) + V_1 + R_4 \cdot (I_C - I_B) = 0 & (\text{malha } C) \\ R_2 \cdot (I_B - I_A) + R_4 \cdot (I_B - I_C) - V_{I_1} = 0 & (\text{malha } B) \end{cases}$$

$P < 0 \Rightarrow$ fonte

Fonte I_1

$P_{V_{I_1}} = V_{I_1} \times (-I_B)$

$= 19 \times (5m)$

$= -95mW$

Mas $I_B = I_1 \Rightarrow$ logo $V_{I_1} = ?$

$$\begin{cases} I_A \cdot (R_1 + R_3 + R_2) + I_B \cdot (-R_2) + I_C \cdot (-R_3) = 0 \\ I_A \cdot (-R_3) + I_B \cdot (-R_4) + I_C \cdot (R_3 + R_4) = -V_1 \\ I_A \cdot (-R_2) + I_B \cdot (R_2 + R_4) + I_C \cdot (-R_4) = V_{I_1} \end{cases}$$

V_{I_1} ↓ INCÓGNITA

$I_A = 3.5mA$

$I_C = 4mA$

$V_{I_1} = 19V$

$$I_A \cdot (R_1 + R_2 + R_3) + I_C \times (-R_3) + 0 = R_2 \cdot I_B$$

$$I_A \cdot (-R_3) + I_C \cdot (R_3 + R_4) + 0 = -V_1 + I_B \cdot R_4 \Rightarrow$$

$$I_A \cdot (-R_2) + I_C \cdot (-R_4) - V_{I_1} = -I_B \cdot (R_2 + R_4)$$

I.

$$\begin{bmatrix} (R_1+R_2+R_3) & (-R_3) & 0 \\ (-R_3) & (R_3+R_4) & 0 \\ (-R_2) & (-R_4) & -1 \end{bmatrix} \begin{bmatrix} I_A \\ I_C \\ V_{I_1} \end{bmatrix} = \begin{bmatrix} R_2 \cdot I_B \\ -V_1 + I_B \cdot R_4 \\ -I_B \cdot (R_2 + R_4) \end{bmatrix}$$

$$\begin{cases} -I_B - I_C + 0 + 0 + 0 + 0 = \cancel{0} - I_A \\ 0 + I_C - I_D - I_E + 0 + 0 = 0 \\ I_B + 0 + 0 + I_E - I_F + 0 = 0 \\ 0 + R_2 I_C + R_4 I_D + 0 + 0 - V_{I_1} = 0 \\ 0 + 0 - R_4 I_D + R_3 I_E + 0 + 0 = -V_1 \\ R_1 I_B - R_2 I_C + 0 - R_3 I_E + 0 + 0 = 0 \end{cases}$$

$$\begin{bmatrix} -1 & -1 & 0 & 0 & 0 & 0 \\ 0 & 1 & -1 & -1 & 0 & 0 \\ 1 & 0 & 0 & 1 & -1 & 0 \\ 0 & R_2 & R_4 & 0 & 0 & -1 \\ 0 & 0 & -R_4 & R_3 & 0 & 0 \\ R_1 & -R_2 & 0 & -R_3 & 0 & 0 \end{bmatrix} \begin{bmatrix} I_B \\ I_C \\ I_D \\ I_E \\ I_F \\ V_{I_1} \end{bmatrix} = \begin{bmatrix} \cancel{0} - I_A \\ 0 \\ 0 \\ 0 \\ -V_1 \\ 0 \end{bmatrix}$$

$$\Downarrow$$

$$\begin{cases} I_B = 3.5\,mA \\ I_C = 1.5\,mA \\ I_D = 1.0\,mA \\ I_E = 2.5\,mA \\ I_F = 4,0\,mA \\ V_{I_1} = 19\,V \end{cases}$$

$$P_{V_{I_1}} = V_{I_1} \times (-I_A) = 19 \times (-5m) = -0,095\,W = -95\,mW$$

$$\Downarrow$$

$P < 0 \Rightarrow$ FONTE FORNECE

$I_1 \Rightarrow$ FORNECE ENERGIA ($P < 0$)

c)

$P_{V_1} = 95mW \Rightarrow$ ENERGIA FORNECIDA

AS RESISTÊNCIAS CONSOMEM ENERGIA

- $P_{R_1} = R_1 I_1^2$
- $P_{R_2} = R_2 (I_A - I_B)^2$
- $P_{R_3} = R_3 \cdot (I_A - I_C)^2$
- $P_{R_4} = R_4 \cdot (I_B - I_B)^2$

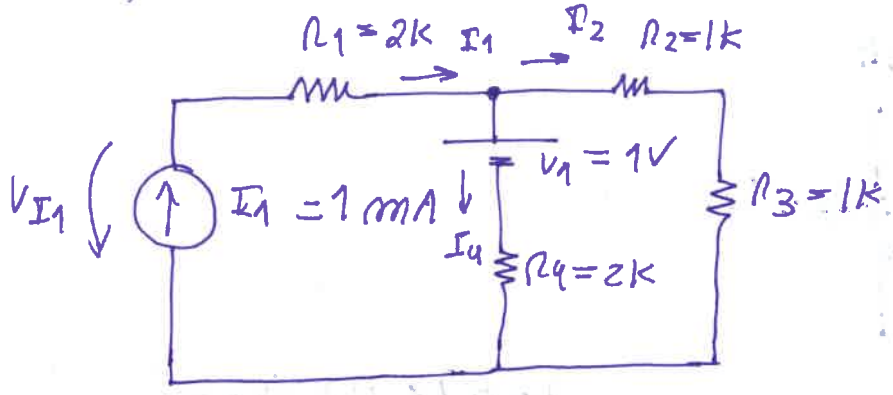
- A FONTE DO CIRCUITO CONSOME POIS $P_{V_1} = V_1 \times I_C \geq 0$

d)

SIMULAÇÃO

9)

5



a) $P_{V_{I1}} = V_{I1} \times (-I_1)$

- $I_1 = I_2 + I_4$ $\quad I_1 = 1mA \Rightarrow V_{I1} = ?$

$\Rightarrow$

- $-V_{I1} + R_1 I_1 + V_1 + R_4 I_4 = 0$
- $R_2 I_2 + R_3 I_2 - R_4 I_4 - V_1 = 0$

$$I_2 + I_4 + 0 = 0 + I_1$$
$$0 + R_4 I_4 - V_{I1} = -R_1 I_1 - V_1$$
$$(R_2 I_2 + R_3 I_2) + (-R_4 I_4) + 0 = V_1$$

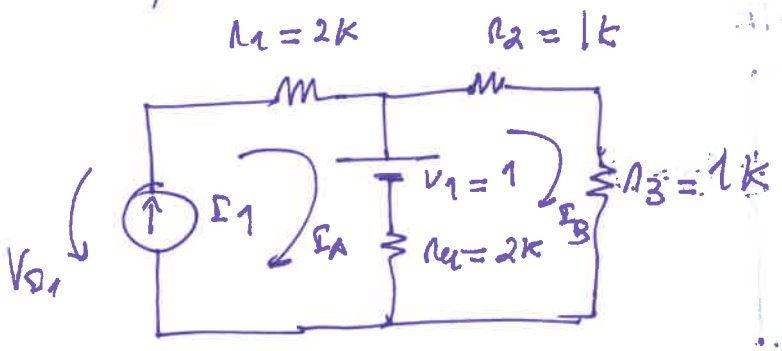
$$\begin{bmatrix} 1 & 1 & 0 \\ 0 & R_4 & -1 \\ (R_2+R_3) & -R_4 & 0 \end{bmatrix} \begin{bmatrix} I_2 \\ I_4 \\ V_{I1} \end{bmatrix} = \begin{bmatrix} +I_1 \\ -R_1 I_1 - V_1 \\ V_1 \end{bmatrix}$$

$I_2 = +0.75mA$

$I_4 = +0.25mA$

$P_{I_1} = -3.5mW \Rightarrow$ I_1 ΠΑΡΕΧΕΙ ΕΝΕΡΓΕΙΑ

b)



$D_A = I_1 \Rightarrow V_{D1} = ?$

$$\begin{cases} -V_{I1} + R_1 I_A + V_1 + R_4 \cdot (I_A - I_B) = 0 \\ R_2 I_B + R_3 I_B + R_4 \cdot (I_B - I_A) - V_1 = 0 \end{cases}$$

$$\begin{cases} -V_{I1} + R_1 I_A + V_1 + R_4 D_A - R_4 D_B = 0 \\ R_2 I_B + R_3 I_B + R_4 D_B - R_4 I_A - V_1 = 0 \end{cases}$$

$P_{V1} = V_1 \times (I_B - I_A)$

$$-R_4 I_B - V_{D1} = -I_A \times (R_1 + R_4) - V_1$$

$$D_B (R_2 + R_3 + R_4) + 0 = R_4 I_A + V_1$$

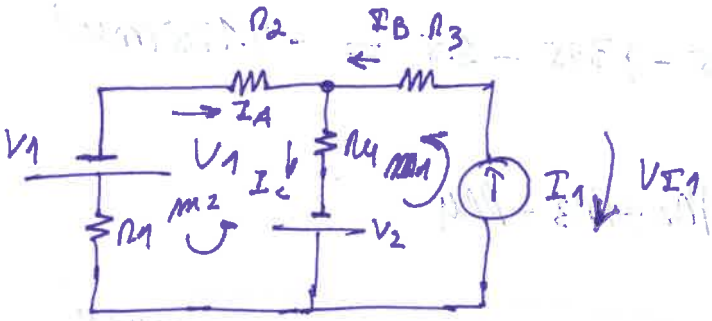
$$\begin{bmatrix} -R_4 & -1 \\ (R_2+R_3+R_4) & 0 \end{bmatrix} \begin{bmatrix} I_B \\ V_{I1} \end{bmatrix} = \begin{bmatrix} -I_A \times (R_1+R_4) - V_1 \\ R_4 \times I_A + V_1 \end{bmatrix}$$

$I_B = 9{,}5\,mA$

$P_{I1} = -3{,}5\,mW$

10) a)

7



$V_1 = 2\,V$

$V_2 = 1\,V$

$I_1 = 1\,mA$

$R_1 = R_2 = R_3 = 2k$

$R_4 = 4k$

$I_A + I_B = I_C$

$-V_{I1} + R_3 I_B + R_4 I_C - V_2 = 0$

$-R_2 I_A - V_1 + R_1(-I_A) + V_2 - R_4 I_C = 0$

$I_B = I_1$

$I_A - I_C + 0 = -I_B$

$0 + R_4 I_C + R_3 I_B - V_{I1} = 0 \quad + V_2$

$-R_2 I_A - R_1 I_A - R_4 I_C + 0 + 0 = V_1 - V_2$

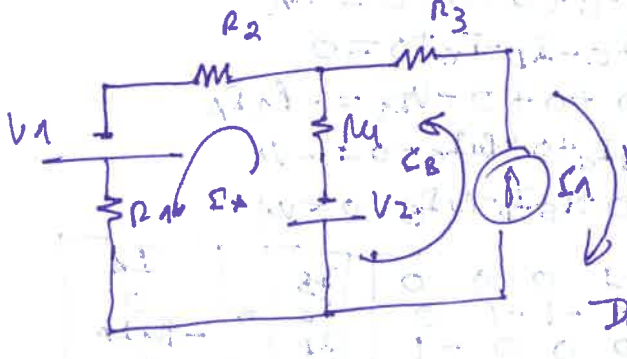
$$\begin{bmatrix} 1 & -1 & 0 \\ 0 & R_4 & -1 \\ -(R_2+R_1) & -R_4 & 0 \end{bmatrix} \begin{bmatrix} I_A \\ I_C \\ V_{I1} \end{bmatrix} = \begin{bmatrix} -I_B \\ -R_3 I_B + V_2 \\ V_1 - V_2 \end{bmatrix}$$

$P_{V1} = V_1 \cdot I_A = -1.25mW$

$P_{V2} = V_2 \cdot (-I_C) = -9.375mW$

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b)



$R_2 I_A - V_1 + R_1 I_A + V_2 + R_4 (I_A - I_B) = 0$

$I_B = I_1$

$V_{I1} = R_3 I_B + R_4 \times (I_B - I_A) - V_2$

$$I_A = \frac{V_1 - V_2 + R_4 I_B}{R_2 + R_1 + R_4}$$

$P_{VI1} = I_A \cdot (-V_{I1}) = -2.5mW$

Potência fornecida $= P_{V1} + P_{R2} + P_{I1} =$

$= -1.25 - 9.375 - 2.5 \approx -9.125\ mW$

Potência consumida $= P_{R1} + P_{R2} + P_{R3} + P_{R4}$

$P_{R1} = I_1^2 \times R_1$

$P_{R2} = R_3 \times I_B^2$

$P_{R3} = R_3 \times I_B^2$

$P_{R4} = R_4 \times (I_B - I_A)^2$

$\approx 9.125\ mW$

Potência consumida

$P_{R1} = R_1 I_A^2$

$P_{R2} = R_2 I_B^2$

$P_{R3} = R_3 I_C^2$

$P_{R4} = R_4 I_E^2$

$P_{R5} = R_5 I_D^2$

$\approx 13.5\ mW$

11) a) $P_{D1} = -13.4\ mW \Rightarrow$ fornece

$P_{V1} = -908\ mW \Rightarrow$ fornece

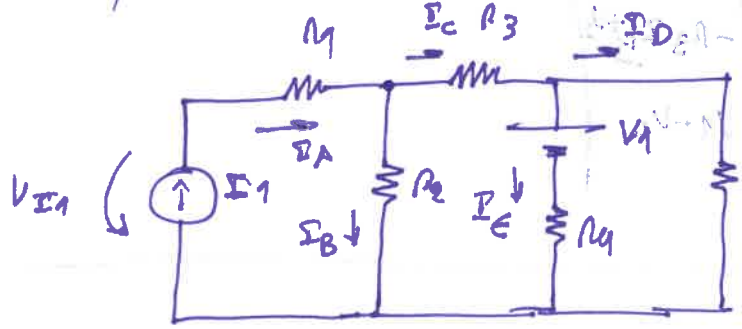
$\approx 13.5\ mW$

$V_1 = 2V$

$D_1 = 1mA$

$R_1 = 10k$

$R_2 = R_3 = R_4 = R_5 = 5k$



- $I_A = I_1$
- $I_A = I_B + I_C$
- $I_C = I_D + I_E$
- $-V_{I1} + R_1 I_A + R_2 I_B = 0$
- $R_3 I_C + V_1 + R_4 I_E - R_2 I_B = 0$
- $R_5 I_D - R_4 I_E - V_1 = 0$

$\Rightarrow$

$I_B + I_C + 0 + 0 + 0 = I_A$

$0 + I_C - I_D - I_E + 0 = 0$

$R_2 I_B + 0 + 0 + 0 - V_{I1} = -R_1 I_A$

$-R_2 I_B + R_3 I_C + 0 + R_4 I_E + 0 = -V_1$

$0 + 0 + R_5 I_D - R_4 I_E + 0 = V_1$

$$\begin{bmatrix} 1 & 1 & 0 & 0 & 0 \\ 0 & 1 & -1 & -1 & 0 \\ R_2 & 0 & 0 & 0 & -1 \\ -R_2 & R_3 & 0 & R_4 & 0 \\ 0 & 0 & R_5 & -R_4 & 0 \end{bmatrix} \begin{bmatrix} I_B \\ I_C \\ I_D \\ I_E \\ V_{I1} \end{bmatrix} = \begin{bmatrix} I_A \\ 0 \\ -R_1 I_A \\ -V_1 \\ V_1 \end{bmatrix}$$